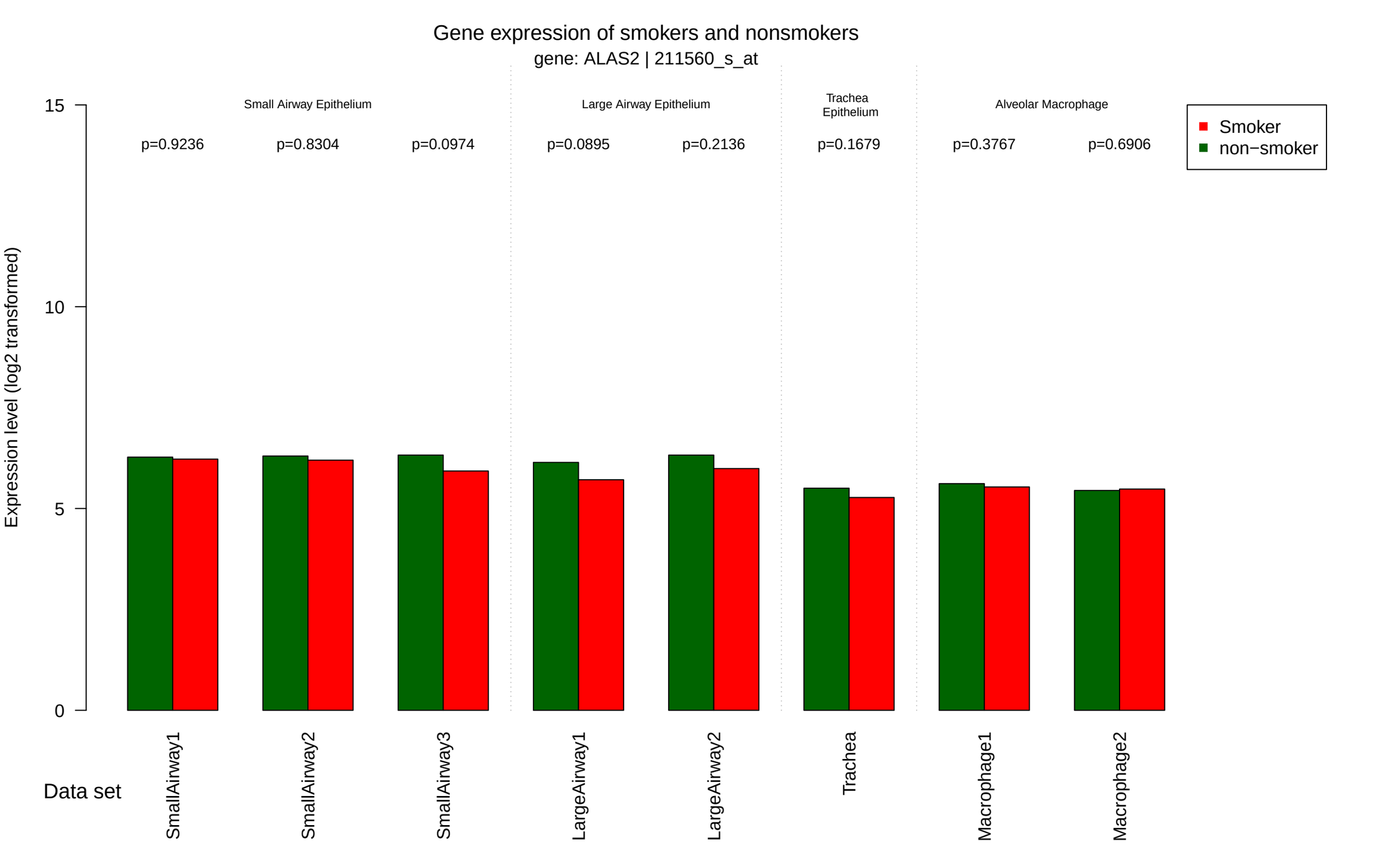
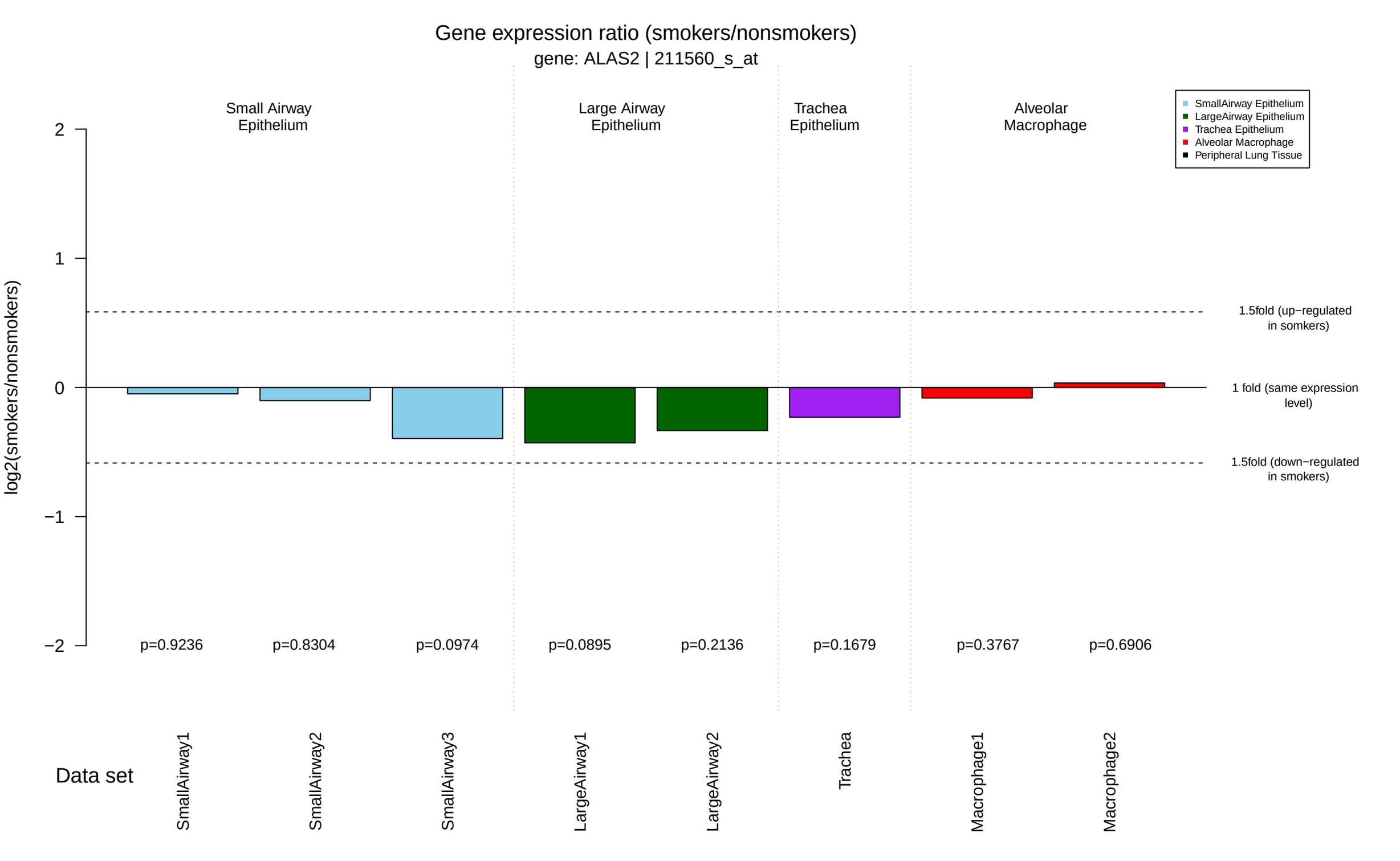
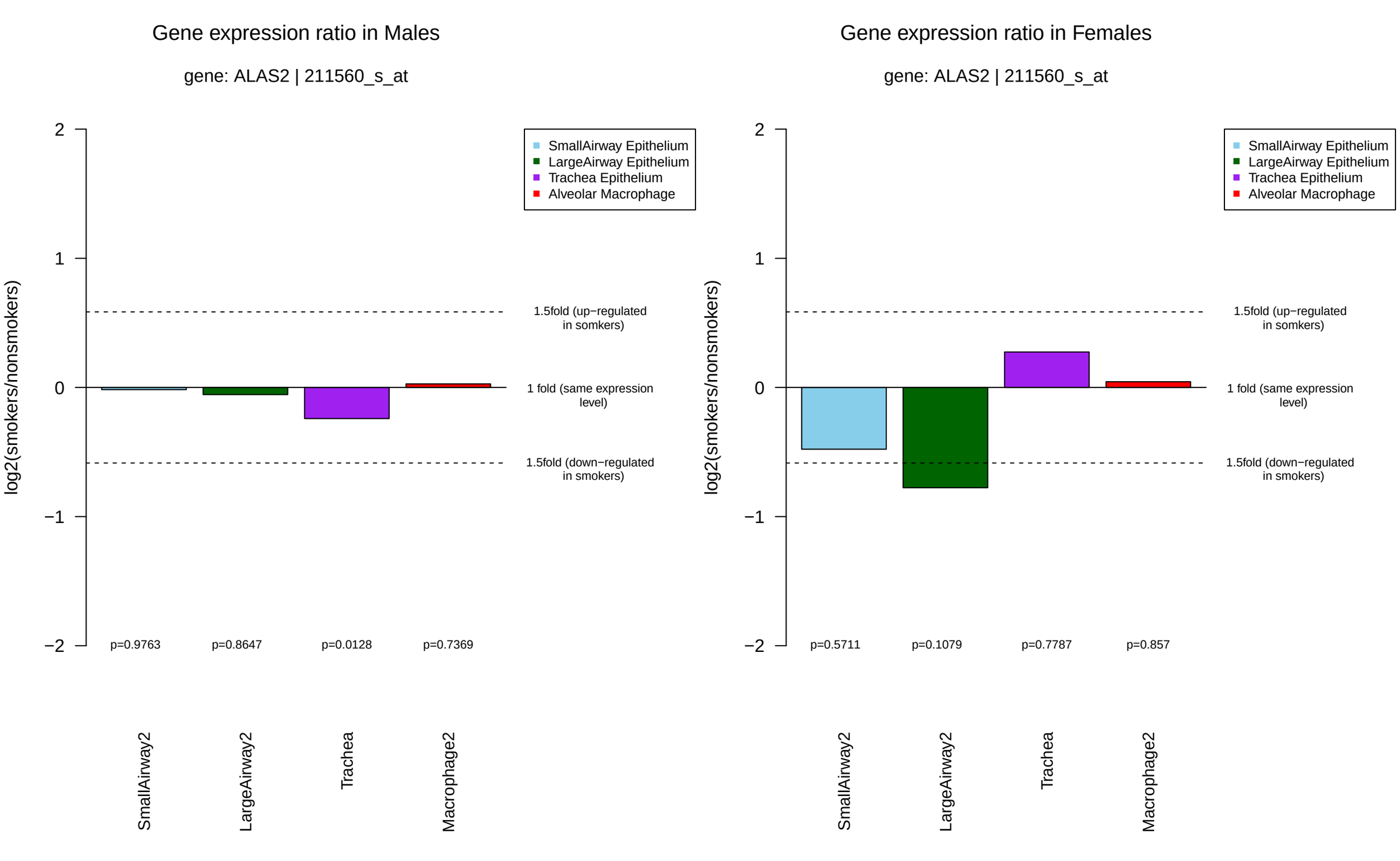
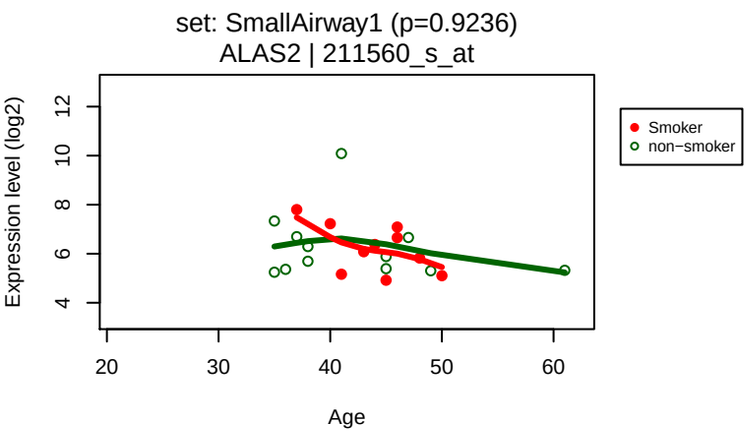
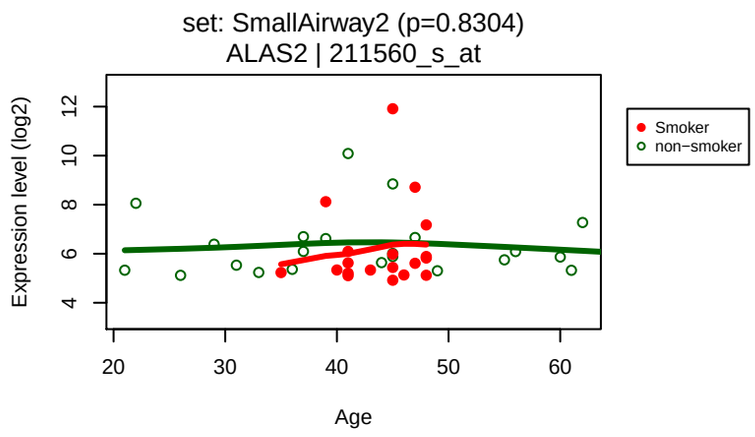
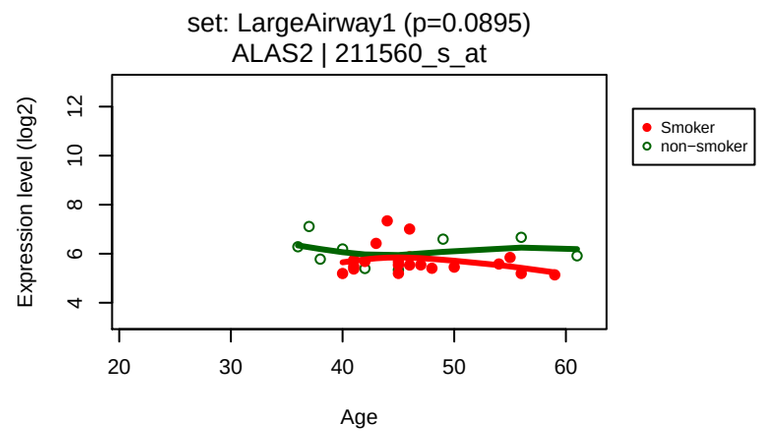

Gene expression of smokers and nonsmokers
gene: ALAS2 | 211560_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9236
p=0.8304
p=0.0974
p=0.0895
p=0.2136
p=0.1679
p=0.3767
p=0.6906
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

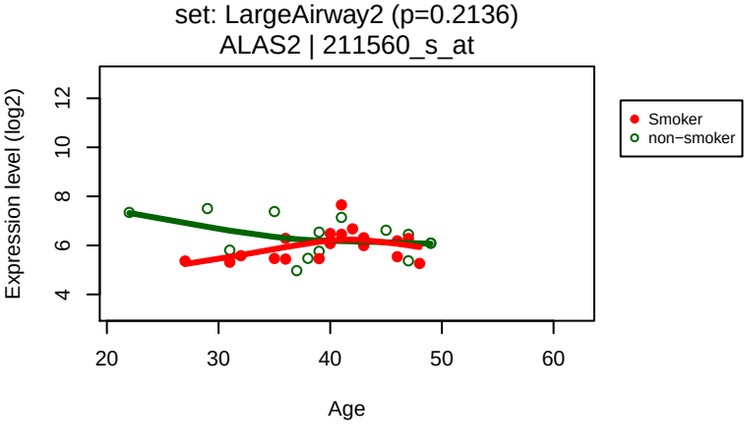

Gene expression ratio (smokers/nonsmokers)
gene: ALAS2 | 211560_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9236
p=0.8304
p=0.0974
p=0.0895
p=0.2136
p=0.1679
p=0.3767
p=0.6906
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

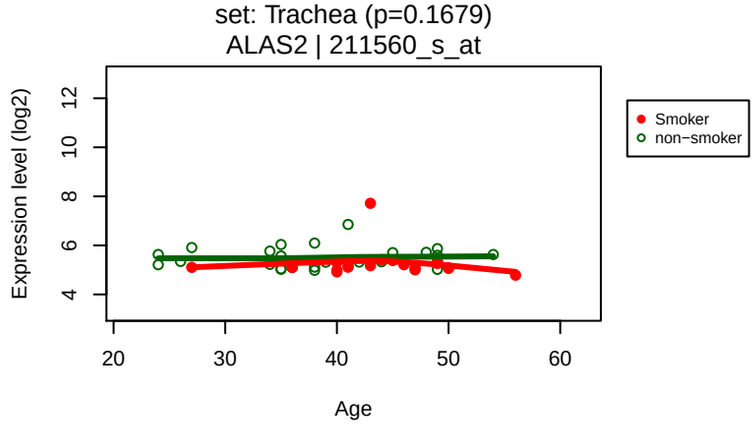

Gene expression ratio in Males
gene: ALAS2 | 211560_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9763
p=0.8647
p=0.0128
p=0.7369
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALAS2 | 211560_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5711
p=0.1079
p=0.7787
p=0.857
SmallAirway2
LargeAirway2
Trachea
Macrophage2

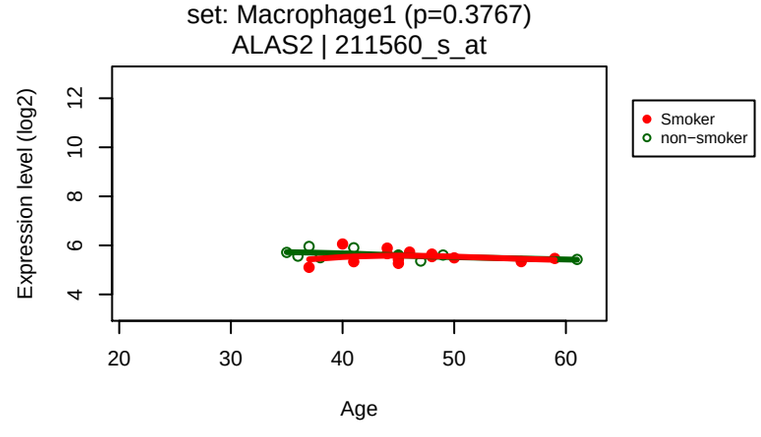

set: SmallAirway1 (p=0.9236)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker

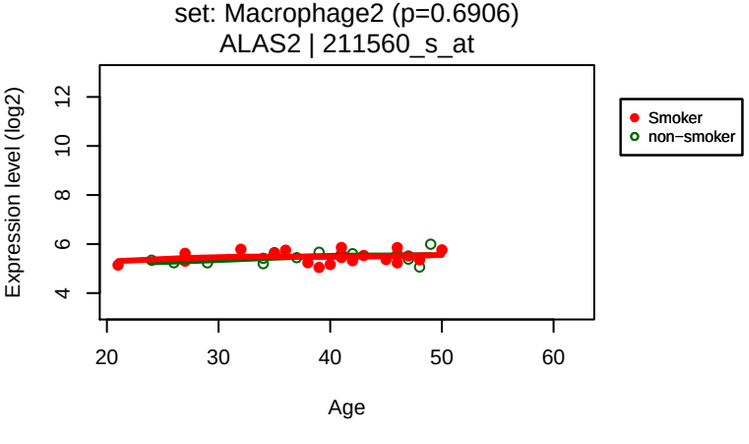

set: SmallAirway2 (p=0.8304)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker

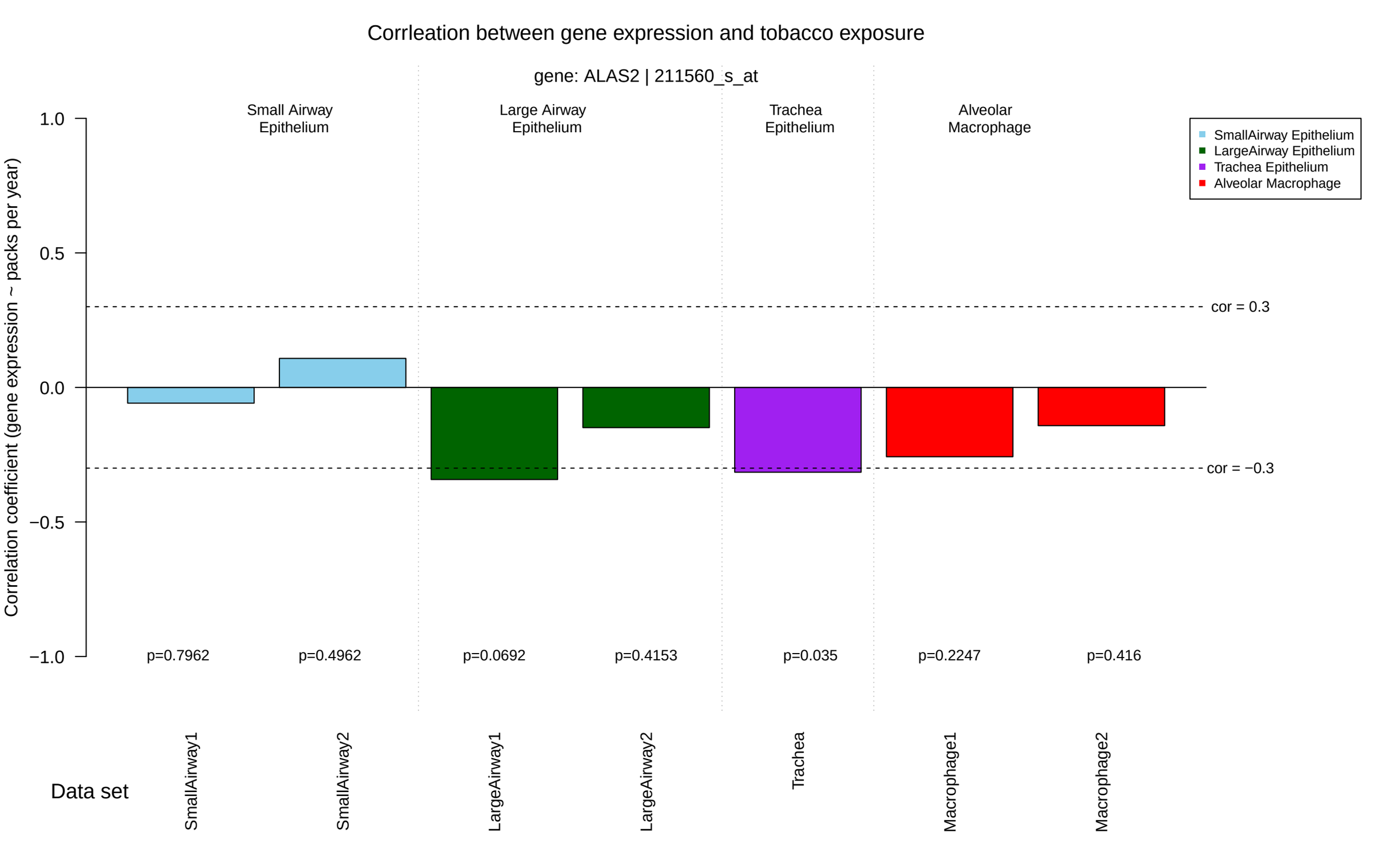

set: LargeAirway1 (p=0.0895)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2136)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1679)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3767)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6906)
ALAS2 | 211560_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALAS2 | 211560_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7962
p=0.4962
p=0.0692
p=0.4153
p=0.035
p=0.2247
p=0.416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2